

Olanrewaju Muili

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Researcher in reinforcement learning, observation reliability, and reliable AI systems. Recent first-author ICLR 2027 submission on reliability estimation for RL agents under observation corruption.

Research Interests

Reinforcement learning, reliable and robust AI, observation reliability, adversarial robustness, and sequential decision-making under corrupted or uncertain information. Current work focuses on estimating whether individual observations can be trusted from observation–action histories, understanding the limits of such estimation, and developing agents that can use reliability signals during decision-making.

Education

University of Colorado Boulder

Oct 2025

M.S., Computer Science, GPA 3.97/4.0

Georgia State University

May 2022

M.S., Geosciences, GPA 3.91/4.0

Selected Research

Dynamics-Consistent Observation Reliability in Reinforcement Learning

Aug 2026 – Present

Lead Researcher / First Author

- Developed a per-step, per-channel framework for estimating whether an RL agent’s observations can be trusted under sensor corruption, with evaluation across nine fault families, four MuJoCo environments, five seeds, and closed-loop rollouts.
- Established theoretical limits of surprise-based and history-only reliability estimation, including conditions under which plausible sensor faults cannot be distinguished from clean observations.
- Introduced Dynamics-Consistent Reliability (DCR), combining a learned one-step dynamics model with two-sided, multi-scale residual statistics; achieved held-out AUROC of 0.994, 0.986 and 0.995 on Hopper, Walker2d and HalfCheetah.
- Evaluated reliability-aware control by replacing flagged observations with dynamics-model predictions, recovering 47–65% of lost return on HalfCheetah while identifying fault-isolation failures on Walker2d.
- Manuscript: *Surprise Is Not Enough: Dynamics-Consistent Observation Reliability in Reinforcement Learning*, under review at ICLR 2027.

BASE AI Catastrophic Risk Fellowship

Oct 2026 – Dec 2026

Incoming Research Fellow / Mentor: Sijia Liu (Michigan State University & MIT-IBM Watson AI Lab, IBM Research)

- Studying whether machine unlearning and safety alignment transfer across foundation-model families, modalities and distillation stages; building the benchmark suite and transfer-enhancing objectives grounded in robust optimization and invariance learning.

GuardScope: Layered Prompt-Injection Guardrail Evaluation

2026

Researcher

- Measured what layered prompt-injection guardrails detect under matched adversarial transformations of a 625-prompt benchmark over 18 attack categories; manuscript in preparation for IEEE SaTML 2027.

Georgia State University

2020 – 2022

Graduate Researcher, Geosciences (with Dr. Hassan A. Babaie)

- Machine-learning susceptibility modelling for sinkhole hazards in west central Florida; published in Applied Computing and Geosciences (2025).

Publications and Manuscripts

Peer-Reviewed Publications

- **Muili, O.** and Babaie, H. A. (2025). *Sinkhole susceptibility analysis using machine learning for west central Florida*. Applied Computing and Geosciences.

Manuscripts & Preprints

- **Muili, O.** et al. *Surprise Is Not Enough: Dynamics-Consistent Observation Reliability in Reinforcement Learning*. Under review, ICLR 2027.
- **Muili, O.** et al. *GuardScope: Measuring What Layered Prompt-Injection Guardrails Detect Under Matched Adversarial Transformations*. In preparation for IEEE SaTML 2027.
- **Muili, O.** (2026). *Evaluating hybrid guardrail architectures for prompt injection defense in large language models*. SSRN preprint, DOI 10.2139/ssrn.6246379.

Professional Experience

Handshake

Feb 2026 – Present

AI Safety Evaluator / Red Team Contributor (Contract)

- Sycophancy and behavioural-manipulation evaluations of frontier models over single- and multi-turn conversations; adversarial prompt design; rubric labels and comparative judgments for offline benchmarking.

Deep Anchor Solutions

Oct 2025 – Sept 2026

Research Engineer, U.S. DOE Marine Energy Fellowship

- Scripted 2-D axisymmetric finite-element pipeline for deeply embedded ring-anchor pullout, matching published capacity within 2.1% across 40+ automated configurations.

The Doe Run Company, Viburnum, MO

Jun 2022 – Sept 2025

Data Geoscientist and ML Researcher

- Production 3-D mineral-exploration ML pipelines (CNN, GNN, random forest, XGBoost) with spatial cross-validation and probability calibration; contributed to a workflow that cut exploration-targeting effort by about 38%.

Technical Skills

Methods: model-based and model-free RL (SAC, PPO), statistical hypothesis testing and calibration, anomaly and OOD detection, probabilistic modelling, adversarial robustness, benchmark and protocol design

Tools: Python, PyTorch, Hugging Face, MuJoCo/Gymnasium, NVIDIA Isaac Lab, NumPy/pandas, LaTeX, Docker, Kubernetes, AWS/GCP

Awards, Fellowships and Service

- BASE AI Catastrophic Risk Fellowship, 2026
- U.S. DOE Marine Energy Postgraduate Fellowship, 2025–2026
- Presidential AI Challenge, State Competition Judge, 2026